Input
Processing
Output
Data I/O
Process
Decision
Database
Terminal

Metagenome assemblies
FASTA + sample IDs / manifest

Preprocessing
Validate and filter contigs
Normalize IDs; preserve ID mapping

Contig quality control
Sequence statistics; prepared contigs

Reads
Paired / interleaved FASTQs

Map reads to prepared contigs
CoverM
Coverage and mapping BAM

Yes

Reads?

No

Feature extraction
Predict CDS, rRNA and tRNA
pProdigal · barrnap · ptRNAscan-SE

Prepare feature sequences
CDS / proteins and coordinates
Filter protein sequences

Annotation
Search protein / rRNA references
FAST / BLAST+; apply hit thresholds

MPDB references
Prepared with build_db

Integrate feature annotations
Functional IDs; per-DB taxonomy / LCA
MetaPathways · BEDTools

Count and normalize features
SAMtools · featureCounts · MP
Counts · RPKM · TPM

Export annotated features
Tables · GFF / GenBank
Sequence-backed Pathway Tools inputs

Coverage / feature abundance

Contig-to-genome map

MAGs?

Yes

No

Partition annotated features
MAGSplitter
Existing features + genome sequences

No

Infer pathways?

Yes

Licensed
Pathway Tools

Community pathway inference
Build PGDB · Pathway Tools / PathoLogic
Compatibility filter; taxonomic pruning

Build population PGDBs
Pathway Tools / PathoLogic

Extract pathways and ORF links
Camelot · MetaPathways

Community [population] PGDBs

Index all available results
Linked results schema · report / explorer

MP Explorer
Search · subset · export